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Radian Measure

Two formulas, $s = r\theta$ and $A = \frac{1}{2}r^2\theta$. Every question is a shape built from sectors and triangles — break it into pieces you know.

THE TOPIC ON ONE MAP

Radian Measure

01 — Radians, arcs and sectors

$s = r\theta$

$A = \frac{1}{2}r^2\theta$

in terms of π

02 — Triangles and segments

segment = sector – triangle

obtuse angle

03 — Composite shapes

total area

perimeter

04 — Equations in r and θ

area and perimeter given

quadratic in r

ONLY IN RADIAN

$$\text{arc } s = r\theta \qquad \text{sector } A = \frac{1}{2}r^2\theta \qquad \text{triangle}$$

$$\frac{1}{2}ab \sin C$$

$\pi \text{ rad} = 180^\circ$ • calculator in RAD mode

01 Radians, Arcs and Sectors

$\pi = 180^\circ$. With θ in radians, the arc and the sector are one multiplication each.

WHEN YOU SEE → YOU DO

“exact value”, “in terms of π ”	→ keep π and surds
“perimeter of the sector”	→ $2r + r\theta$ — two radii plus the arc
“area of the sector is 15, find r ”	→ $\frac{1}{2}r^2\theta = 15$
a major (reflex) sector	→ angle = $2\pi - \theta$

WORKED MODEL • OCTOBER 2019

Sector, $\theta = 1.25$, area 15 cm^2 .

1 $\frac{1}{2}r^2(1.25) = 15 \Rightarrow r^2 = 24 \Rightarrow r = 2\sqrt{6}$

area formula

2 $P = 2r + r\theta = 4\sqrt{6} + 2.5\sqrt{6}$

perimeter

ANSWER $r = 2\sqrt{6}$, $P = \frac{13}{2}\sqrt{6}$

× **TRAP** Calculator in DEG mode gives wrong sines. Check: $\sin 1.2 = 0.932$ in RAD.

IDEA 01

Practise: Radians, Arcs and Sectors

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Arc and sector

P1 · TOPIC 10 3 marks

A sector has radius 6 cm and angle $\frac{2\pi}{3}$. Find, in terms of π , its arc length, area and perimeter.

Q2 DR ESLAM Find θ

P1 · TOPIC 10 2 marks

A sector of radius 5 cm has area 20 cm^2 . Find the angle in radians.

 $\theta =$ _____

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 10

5 marks

1 Without using a calculator, write down:

(i) 60° *in radians*,

(ii) $\sin 45^\circ$

(iii) $\tan \pi$

(iv) $\frac{5\pi}{6}$ *radians in degrees*

(iv) $\cos 0$

(5 marks)

Q4**PRACTICE**

Practice · Easy

P1 · TOPIC 10**4 marks**

- 2 A sector of a circle, OPQ , is such that it has radius 4 cm and the angle at its centre, O , is $\frac{\pi}{4}$ radians.

(i) Find the length of the arc PQ .

(ii) Find the area of the sector OPQ .

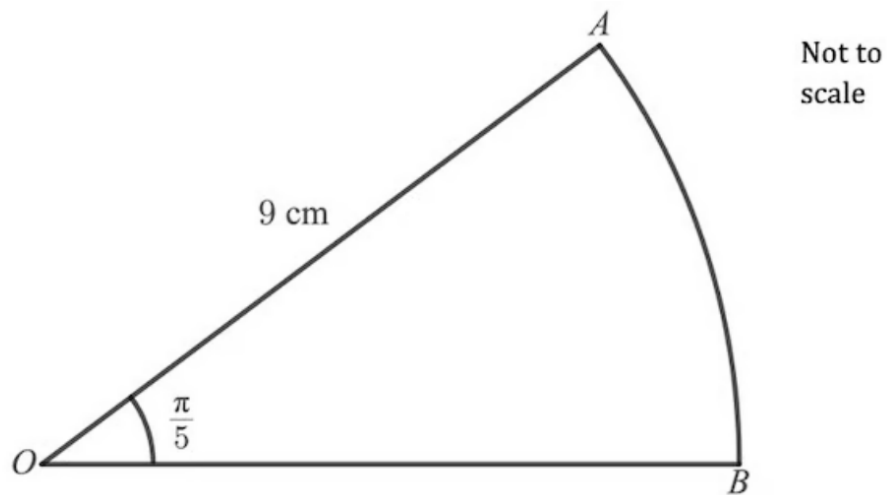
(4 marks)

Q5**PRACTICE**

Practice · Medium

P1 · TOPIC 10 5 marks

3 The sector of a circle is shown below.



Find the area and the perimeter of the sector.

(5 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · October 2019 · Q1

P1 · TOPIC 10 5 marks

1.

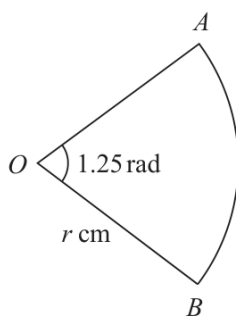
**Figure 1**

Figure 1 shows a sector AOB of a circle with centre O and radius r cm.

The angle AOB is 1.25 radians.

Given that the area of the sector AOB is 15 cm^2

- (a) find the exact value of r , (2)
- (b) find the exact length of the perimeter of the sector. Write your answer in simplest form. (3)

(Total 5 marks)

Q7**PAST PAPER**

Pearson IAL WMA11 · June 2022 · Q8

P1 · TOPIC 10 10 marks

8.

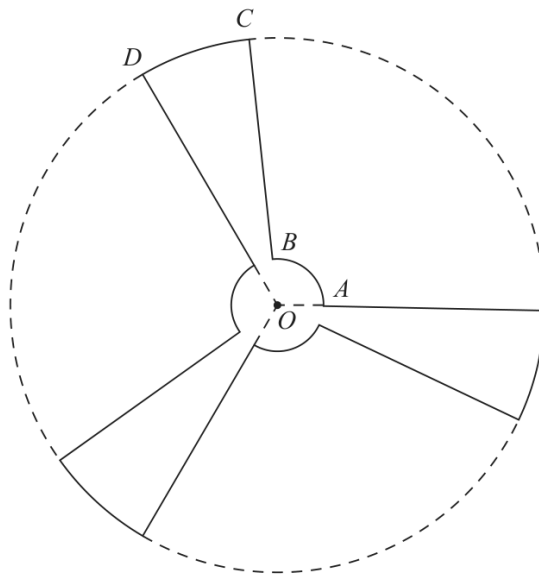
**Figure 3**

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

The fan consists of three identical sections congruent to $OABCD$, shown in Figure 3, where

- $OABO$ is a sector of a circle with centre O and radius 9 cm
- $OBCDO$ is a sector of a circle with centre O and radius 84 cm
- angle $AOD = \frac{2\pi}{3}$ radians

Given that the length of the arc AB is 15 cm,

- (a) show that the length of the arc CD is 35.9 cm to one decimal place. (3)

The face of the fan is modelled to be a flat surface.

Find, according to the model,

- (b) the perimeter of the face of the fan, giving your answer to the nearest cm, (2)
- (c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear. (5)

(Total 10 marks)

Q8 PAST PAPER Pearson IAL WMA11 · October 2024 · Q5

P1 · TOPIC 10 9 marks

5. A plot of land OAB is in the shape of a sector of a circle with centre O .

Given

- $OA = OB = 5 \text{ km}$
- angle $AOB = 1.2 \text{ radians}$

(a) find the perimeter of the plot of land.

(2)

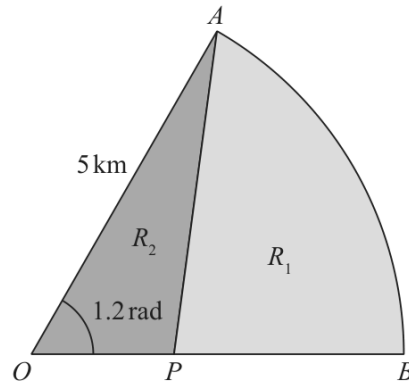


Figure 2

Diagram **NO**
accurately dr

A point P lies on OB such that the line AP divides the plot of land into two regions R_1 and R_2 as shown in Figure 2.

Given that

$$\text{area of } R_1 = 3 \times \text{area of } R_2$$

(b) show that the area of $R_2 = 3.75 \text{ km}^2$

(3)

(c) Find the length of AP , giving your answer to the nearest 100 m.

(4)

(Total for Question 5 is 9 marks)

02 Triangles and Segments

Most shapes add or subtract a triangle. Find its missing side or angle with the sine or cosine rule.

WHEN YOU SEE → YOU DO

“segment”, “shaded region between chord and arc”

$$\longrightarrow \text{sector} - \text{triangle} = \frac{1}{2}r^2(\theta - \sin \theta)$$

two sides and the angle between

$$\longrightarrow \text{cosine rule for the third side} \cdot \text{area } \frac{1}{2}ab \sin C$$

a side opposite a known angle

$$\longrightarrow \text{sine rule}$$

“angle OAB is obtuse”, “ $> \frac{\pi}{2}$ ”

$$\longrightarrow \text{sine rule gives the acute one: use } \pi - \alpha$$

“show that angle AOB is 1.231 radians”

$$\longrightarrow \text{give 4 s.f. or more before rounding}$$

WORKED MODEL • JANUARY 2022

Semicircle $r = 3$, $\angle BOA = 0.7$, $BD \parallel AE$.

$$1 \quad \angle BOD = \pi - 2(0.7) = 1.742$$

symmetry

$$2 \quad \text{segment} = \frac{1}{2}(9)(1.742) - \frac{1}{2}(9) \sin 1.742$$

sector - triangle

$$3 \quad BD^2 = 9 + 9 - 18 \cos 1.742 \Rightarrow BD = 4.59$$

cosine rule

$$\text{ANSWER} \quad \text{area } 3.4 \text{ m}^2, \text{ perimeter } 3(1.742) + 4.59 = 9.8 \text{ m}$$

IDEA 02

Practise: Triangles and Segments

My questions first, then selected practice, then real past papers.

Q9**DR ESLAM**

Segment

P1 · TOPIC 10

3 marks

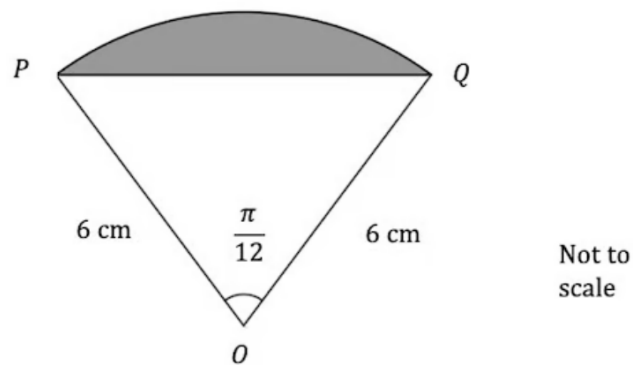
A sector has $r = 4$ cm and $\theta = 1.2$. Find the area of the segment cut off by the chord.

SELECTED PRACTICE

Q10 PRACTICE Practice • Easy

P1 • TOPIC 10 6 marks

- 7 (a) The diagram below shows the sector of a circle, OPQ , radius 6 cm and centre angle $\frac{\pi}{12}$ radians.



Use the formula $\frac{1}{2}ab \sin C$ to find the area of the triangle OPQ .

(2 marks)

- (b) Find the area of the sector OPQ .

(2 marks)

- (c) Hence or otherwise find the area of the shaded segment.

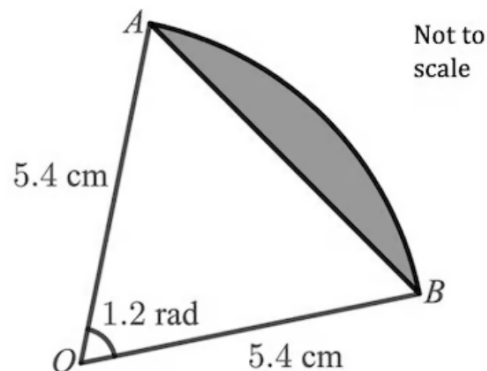
(2 marks)

Q11**PRACTICE**

Practice • Medium

P1 • TOPIC 10 8 marks

5 (a) The diagram below shows the sector of a circle .



- (i) Find the area of the sector OAB , giving your answer to 3 significant figures.
- (ii) Find the area of the triangle OAB , giving your answer to 3 significant figures.
- (iii) Find the area of the shaded segment, giving your answer to 3 significant figures.

(5 marks)

(b) (i) Find the length of the arc AB .

- (ii) Find the perimeter of the sector OAB .

(3 marks)

PAST PAPERS

Q12 PAST PAPER Pearson IAL WMA11 · January 2021 · Q5

P1 · TOPIC 10 9 marks

5.

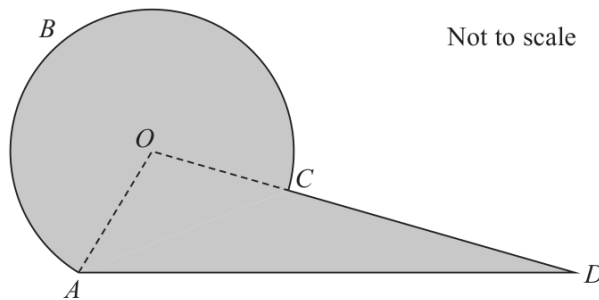


Figure 3

Figure 3 shows the plan view of a viewing platform at a tourist site.

The shape of the viewing platform consists of a sector $ABCOA$ of a circle, centre O , joined to a triangle AOD .

Given that

- $OA = OC = 6$ m
- $AD = 14$ m
- angle $ADC = 0.43$ radians
- angle AOD is an obtuse angle
- OCD is a straight line

find

- (a) the size of angle AOD , in radians, to 3 decimal places, (3)
- (b) the length of arc ABC , in metres, to one decimal place, (2)
- (c) the total area of the viewing platform, in m^2 , to one decimal place. (4)

(Total 9 marks)

Q13 PAST PAPER Pearson IAL WMA11 • October 2021 • Q7

P1 • TOPIC 10 10 marks

7.

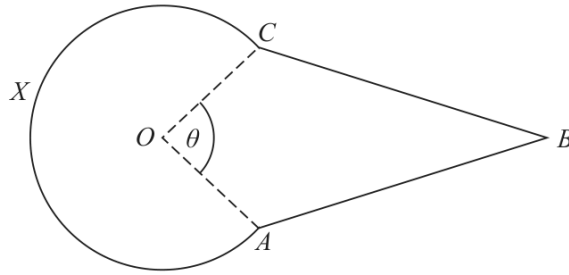
**Figure 3**

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector $OCXA$ of a circle centre O .

In the design

- $OA = OC = 0.6$ m
- $AB = CB = 1.4$ m
- Angle $OAB = \text{Angle } OCB = 2$ radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

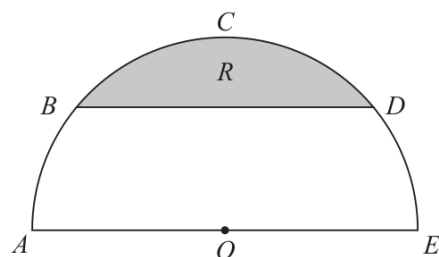
- show that $\theta = 1.64$ radians to 3 significant figures, (4)
- find the perimeter of the sign, in metres to 2 significant figures, (2)
- find the area of the sign, in m^2 to 2 significant figures. (4)

(Total 10 marks)

Q14 PAST PAPER Pearson IAL WMA11 • January 2022 • Q5

P1 • TOPIC 10 7 marks

5.



Not to scale

Figure 2

Figure 2 shows a plan view of a semicircular garden $ABCDEOA$

The semicircle has

- centre O
- diameter AOE
- radius 3 m

The straight line BD is parallel to AE and angle BOA is 0.7 radians.

- (a) Show that, to 4 significant figures, angle BOD is 1.742 radians.

(1)

The flowerbed R , shown shaded in Figure 2, is bounded by BD and the arc BCD .

- (b) Find the area of the flowerbed, giving your answer in square metres to one decimal place.

(3)

- (c) Find the perimeter of the flowerbed, giving your answer in metres to one decimal place.

(3)

(Total 7 marks)

Q15 PAST PAPER Pearson IAL WMA11 · October 2022 · Q8

P1 · TOPIC 10 14 marks

8.

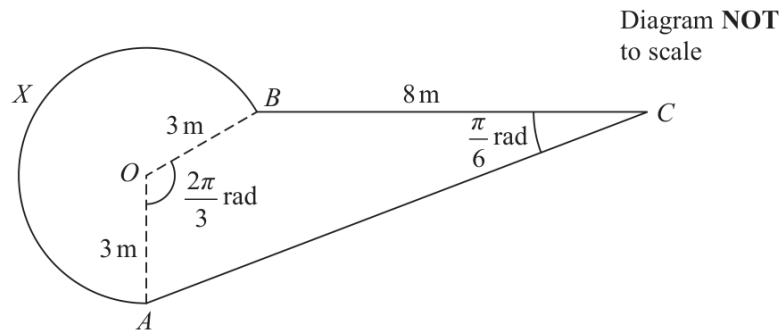
**Figure 2**

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector $AOBX$ of a circle centre O joined to a quadrilateral $AOBC$.

- $BC = 8$ m
- $OA = OB = 3$ m
- angle AOB is $\frac{2\pi}{3}$ radians
- angle BCA is $\frac{\pi}{6}$ radians

(a) Calculate (i) the exact area of the sector $AOBX$,

(ii) the exact perimeter of the sector $AOBX$.

(5)

(b) Calculate the exact area of the triangle AOB .

(2)

(c) Show that the length AB is $3\sqrt{3}$ m.

(2)

(d) Find the total surface area of the pond. Give your answer in m^2 correct to 2 significant figures.

(5)

(Total for Question 8 is 14 marks)

Q16 PAST PAPER Pearson IAL WMA11 • January 2023 • Q6

P1 • TOPIC 10 10 marks

6.

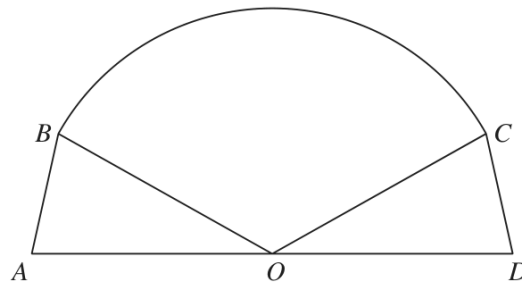
Diagram **NO**
accurately dr**Figure 1**

Figure 1 shows the plan view for the design of a stage.

The design consists of a sector BOC of a circle, with centre O , joined to two congruent triangles OAB and ODC .

Given that

- angle $BOC = 2.4$ radians
- area of sector $BOC = 40 \text{ m}^2$
- AOD is a straight line of length 12.5 m

(a) find the radius of the sector, giving your answer, in m , to 2 decimal places,

(2)

(b) find the size of angle AOB , in radians, to 2 decimal places.

(1)

Hence find

(c) the total area of the stage, giving your answer, in m^2 , to one decimal place,

(3)

(d) the total perimeter of the stage, giving your answer, in m , to one decimal place.

(4)

(Total for Question 6 is 10 marks)

Q17 PAST PAPER Pearson IAL WMA11 · June 2024 · Q5

P1 · TOPIC 10 7 marks

5.

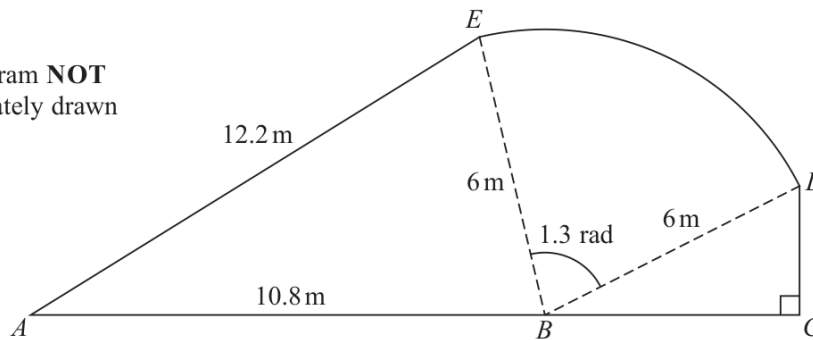
Diagram **NOT**
accurately drawn**Figure 2**

Figure 2 shows the plan view of a garden.

The shape of the garden $ABCDEA$ consists of a triangle ABE and a right-angled triangle BCD joined to a sector BDE of a circle with radius 6 m and centre B .

The points A , B and C lie on a straight line with $AB = 10.8$ m

Angle $BCD = \frac{\pi}{2}$ radians, angle $EBD = 1.3$ radians and $AE = 12.2$ m

- (a) Find the area of the sector BDE , giving your answer in m^2 (2)
- (b) Find the size of angle ABE , giving your answer in radians to 2 decimal places. (2)
- (c) Find the area of the garden, giving your answer in m^2 to 3 significant figures. (3)

(Total for Question 5 is 7 marks)

Q18 PAST PAPER Pearson IAL WMA11 · January 2025 · Q8

P1 · TOPIC 10 13 marks

8.

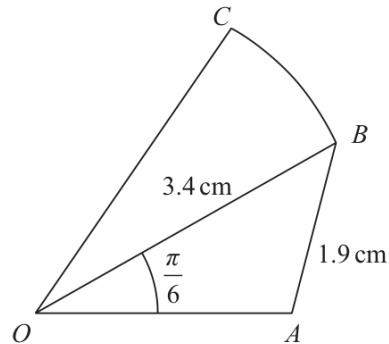
**Figure 1**

Figure 1 shows a sketch of a design for a badge.

The design consists of a triangle OAB joined to a sector OBC of a circle with centre O

In the design

- $OB = 3.4$ cm
- $AB = 1.9$ cm
- angle $AOB = \frac{\pi}{6}$ radians
- angle $OAB > \frac{\pi}{2}$ radians

Making your method clear,

(a) find the size of angle OAB , giving your answer in radians to 4 significant figures, (3)

(b) find the area of triangle OAB , in cm^2 , giving your answer to 3 significant figures. (2)

Given that the ratio of the area of sector OBC to the area of triangle OAB is 3 : 2

(c) show that angle BOC is 0.462 radians to 3 significant figures. (3)

(d) Hence find the perimeter of the badge, in cm, to the nearest integer. (5)

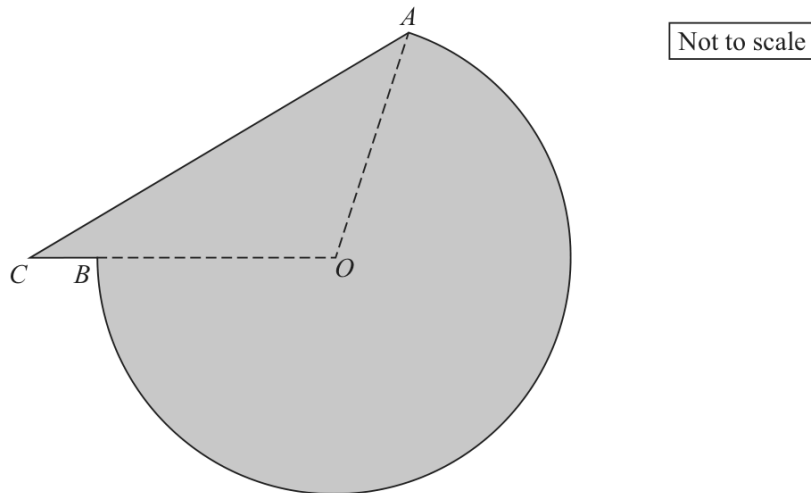
(Total for Question 8 is 13 marks)

Q19**PAST PAPER**

Pearson IAL WMA11 • June 2025 • Q6

P1 • TOPIC 10 9 marks

6.

**Figure 2**

The shaded area in Figure 2 shows the plan view of a helicopter landing pad.

The area consists of the major sector AOB of a circle centre O joined to a triangle AOC .

Given that

- $AO = OB = 15$ m
- $BC = 2$ m
- CBO is a straight line
- angle $ACO = 0.6$ radians

(a) show that angle COA is 1.847 radians to 3 decimal places.

(3)

(b) Find the total area of the helicopter landing pad.
Give your answer in m^2 to 3 significant figures.

(3)

(c) Find the perimeter of the helicopter landing pad.
Give your answer in metres to 3 significant figures.

(3)**(Total for Question 6 is 9 marks)**

Q20**PAST PAPER**

Pearson IAL WMA11 · January 2026 · Q5

P1 · TOPIC 10**9 marks**

5.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

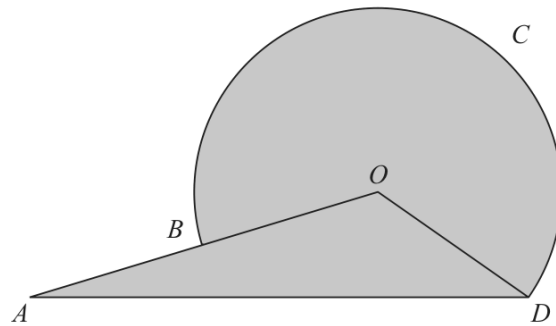


Diagram NOT
to scale

Figure 2

Figure 2 shows the logo for the tourist board of an island.

The logo consists of a sector $BODCB$ of a circle centre O joined to a triangle AOD .

Given that

- ABO is a straight line
- $OD = 3$ cm
- $AD = 8$ cm
- angle $AOD = 2.5$ radians

(a) show that angle ODA is 0.415 radians to 3 significant figures.

(3)

(b) Find the total area of the logo, in cm^2 , to one decimal place.

(3)

(c) Find the perimeter of the logo, $ABCD$, in cm, to one decimal place.

(3)

(Total for Question 5 is 9 marks)

03 Composite Shapes: Break It Up

Gardens, logos, platforms, pools. Label every piece: which sectors, which triangles, which straight edges.

WHEN YOU SEE → YOU DO

- | | |
|---------------------------|---|
| "total area of the ..." | → add the pieces (subtract any hole) |
| "perimeter of the ..." | → walk around the outside only — arcs and straight edges |
| "DA:AO = 3:5", "PQ = 2.8" | → write every radius in terms of one letter |
| "to 3 s.f.", "nearest m" | → keep full accuracy until the last line |

1

Name the pieces

Sector(s), triangle(s), rectangle(s). Mark every radius.

2

Missing lengths and angles

Sine rule, cosine rule, or $\theta = 2\pi - \dots$.

3

Add areas • walk the perimeter

Only the outside edges count for the perimeter.

WORKED MODEL • JANUARY 2024

Sector DOC ($\angle 0.8$, arc 9) joined to major sector $AOBEA$; $DA : AO = 3 : 5$.

1 $OD = \frac{8}{5}AO$, $0.8 \cdot \frac{8}{5}AO = 9 \Rightarrow AO = 7.03$

arc gives the radius

2 $P = 9 + 2(11.25 - 7.03) + 7.03(2\pi - 0.8)$

walk the outside

ANSWER $P = 56.0 \text{ m}$, $A = 186 \text{ m}^2$

IDEA 03

Practise: Composite Shapes

My questions first, then real past papers.

Q21 DR ESLAM Walk the outside

P1 · TOPIC 10 3 marks

A shape is a semicircle of radius 3 cm on top of a 6×4 cm rectangle. Find its perimeter and area.

PAST PAPERS

Q22 PAST PAPER Pearson IAL WMA11 • January 2020 • Q4

P1 • TOPIC 10 9 marks

4.

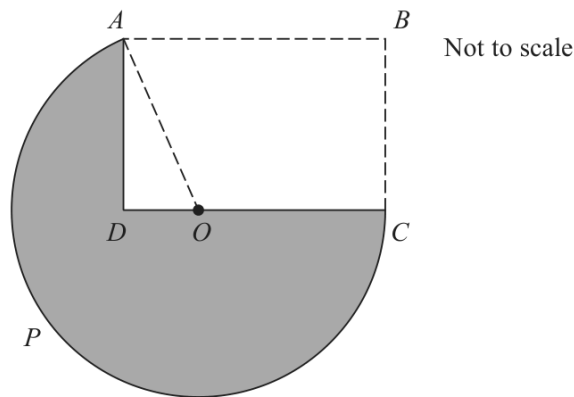


Figure 2

Figure 2 shows the plan view of a house $ABCD$ and a lawn $APCDA$.

$ABCD$ is a rectangle with $AB = 16$ m.

$APCOA$ is a sector of a circle centre O with radius 12 m.

The point O lies on the line DC , as shown in Figure 2.

(a) Show that the size of angle AOD is 1.231 radians to 3 decimal places.

(2)

The lawn $APCDA$ is shown shaded in Figure 2.

(b) Find the area of the lawn, in m^2 , to one decimal place.

(4)

(c) Find the perimeter of the lawn, in metres, to one decimal place.

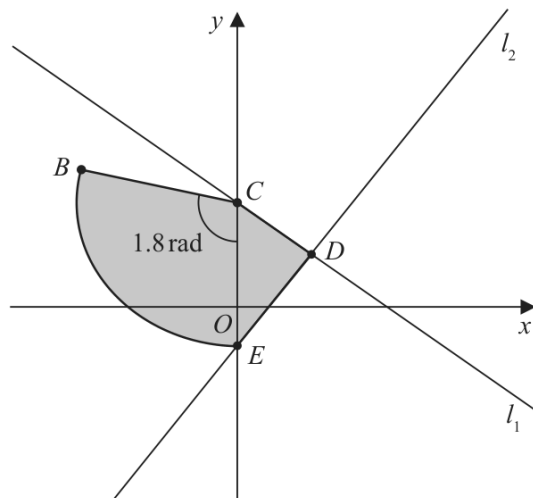
(3)

(Total 9 marks)

Q23 PAST PAPER Pearson IAL WMA11 · June 2021 · Q7

P1 · TOPIC 10 10 marks

7.

**Figure 3**

The line l_1 has equation $4y + 3x = 48$

The line l_1 cuts the y -axis at the point C , as shown in Figure 3.

(a) State the y coordinate of C .

(1)

The point $D(8, 6)$ lies on l_1

The line l_2 passes through D and is perpendicular to l_1

The line l_2 cuts the y -axis at the point E as shown in Figure 3.

(b) Show that the y coordinate of E is $-\frac{14}{3}$

(3)

A sector BCE of a circle with centre C is also shown in Figure 3.

Given that angle BCE is 1.8 radians,

(c) find the length of arc BE .

(3)

The region $CBED$, shown shaded in Figure 3, consists of the sector BCE joined to the triangle CDE .

(d) Calculate the exact area of the region $CBED$.

(3)

(Total 10 marks)

Q24 PAST PAPER Pearson IAL WMA11 · June 2023 · Q5

P1 · TOPIC 10 10 marks

5.

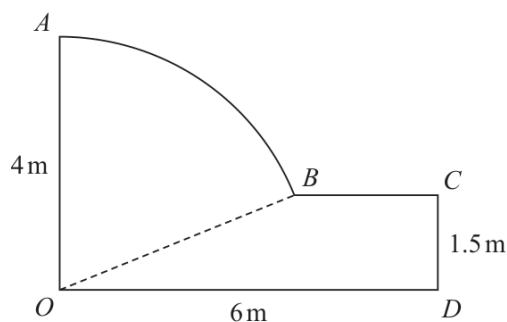
**Figure 1**

Figure 1 shows the plan for a garden.

In the plan

- OA and CD are perpendicular to OD
- AB is an arc of the circle with centre O and radius 4 metres
- BC is parallel to OD
- OD is 6 metres, OA is 4 metres and CD is 1.5 metres

(a) Show that angle AOB is 1.186 radians to 4 significant figures.

(2)

(b) Find the perimeter of the garden, giving your answer in metres to 3 significant figures.

(4)

(c) Find the area of the garden, giving your answer in square metres to 3 significant figures.

(4)

(Total for Question 5 is 10 marks)

Q25 **PAST PAPER** Pearson IAL WMA11 · January 2024 · Q8

P1 · TOPIC 10 **9 marks**

8.

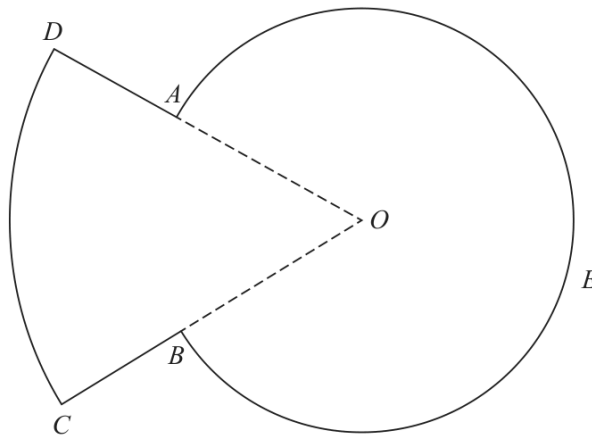


Figure 3

Figure 3 shows a sketch of the plan view of a platform.

The plan view of the platform consists of a sector DOC of a circle centre O joined to a sector $AOBEA$ of a different circle, also with centre O .

Given that

- angle $AOB = 0.8$ radians
- arc length $CD = 9$ m
- $DA : AO = 3 : 5$

(a) show that $AO = 7.03$ m to 3 significant figures.

(3)

(b) Find the perimeter of the platform, in m, to 3 significant figures.

(3)

(c) Find the total area of the platform, giving your answer in m^2 to the nearest whole number.

(3)

(Total for Question 8 is 9 marks)

7.

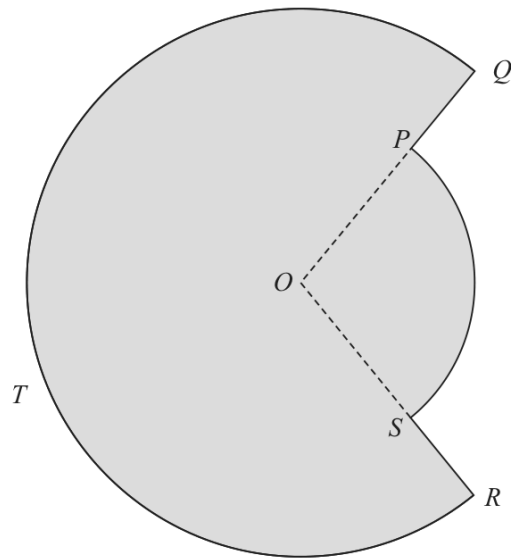


Figure 2

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 2 shows the plan view of a design for a swimming pool.

The design consists of a sector POS of a circle centre O joined to a major sector $QORTQ$ of a different circle, also with centre O .

Given that

- angle POS is 1.65 radians
- the area of sector POS is 30 m^2
- $PQ = 2.8 \text{ m}$

(a) show that, to 3 significant figures, $OQ = 8.83 \text{ m}$

(3)

(b) Find the total surface area of the swimming pool in m^2 to the nearest integer.

(3)

(c) Find the total perimeter of the swimming pool in metres to 2 significant figures.

(3)

(Total for Question 7 is 9 marks)

04 Equations in r and θ

Two facts (area, perimeter, arc) give two equations. Solve them simultaneously — often a quadratic.

WHEN YOU SEE → YOU DO

“the area is 1 and the perimeter is 8”

$$\longrightarrow \frac{1}{2}r^2\theta = 1, 2r + r\theta = 8$$

“show that $5x^2 + 12x - 1500 = 0$ ”

→ area of the big sector – small sector = given

“exact value ... in the form $a + b\sqrt{c}$ ”

→ quadratic formula, no decimals

WORKED MODEL • JUNE 2026

Area 1, perimeter 8, $r > 2$.

$$1 \quad r\theta = 8 - 2r \Rightarrow \frac{1}{2}r(8 - 2r) = 1$$

substitute

$$2 \quad r^2 - 4r + 1 = 0 \Rightarrow r = 2 \pm \sqrt{3}$$

quadratic

ANSWER $r = 2 + \sqrt{3}, \quad \theta = \frac{2}{r^2} = 14 - 8\sqrt{3}$

IDEA 04

Practise: Equations in r and θ

My questions first, then selected practice, then real past papers.

Q27 DR ESAM Two facts, two equations

P1 · TOPIC 10 4 marks

A sector has perimeter 20 cm and area 16 cm^2 . Find the two possible radii.

SELECTED PRACTICE**Q28** PRACTICE Practice · Easy

P1 · TOPIC 10 2 marks

6 The arc length of the sector of a circle is 5 cm .

The radius is twice the length of the arc.

Find the angle at the centre of the circle.

(2 marks)

Q29**PRACTICE**

Practice · Easy

P1 · TOPIC 10**4 marks**

8 The area of a sector is $8\pi \text{ cm}^2$.

The arc length of the sector is $4\pi \text{ cm}$.

Find the radius and the angle at the centre of the circle.

(4 marks)**Q30****PRACTICE**

Practice · Hard

P1 · TOPIC 10**4 marks**

3 A sector of a circle, OPQ , is such that it has radius a cm and the angle at its centre, O , is $a^2 \pi$ radians.

Given that the area of the sector OPQ in cm^2 is three times the length of the arc PQ in cm, find the value of a .

(4 marks)

PAST PAPERS

Q31 PAST PAPER Pearson IAL WMA11 • October 2023 • Q9

P1 • TOPIC 10 7 marks

9.

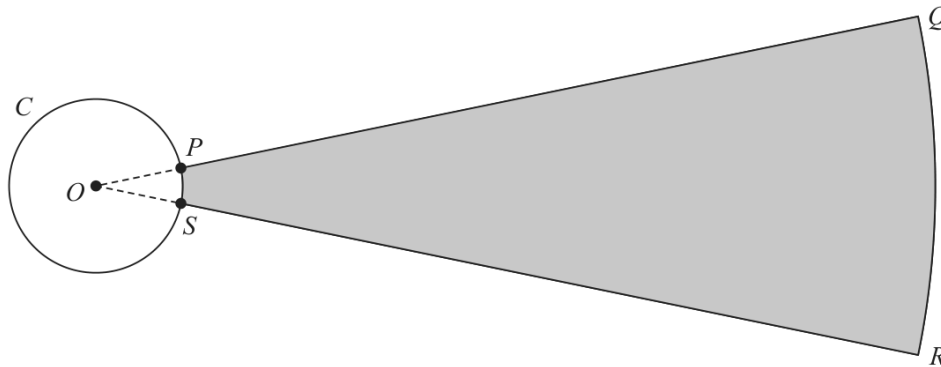
Diagram NO
accurately dr

Figure 3

Figure 3 shows the plan view of the area being used for a ball-throwing competition.

Competitors must stand within the circle C and throw a ball as far as possible into the target area, $PQRS$, shown shaded in Figure 3.

Given that

- circle C has centre O
- P and S are points on C
- $OPQRSO$ is a sector of a circle with centre O
- the length of arc PS is 0.72 m
- the size of angle POS is 0.6 radians

(a) show that $OP = 1.2\text{ m}$

(1)

Given also that

- the target area, $PQRS$, is 90 m^2
- length $PQ = x$ metres

(b) show that

$$5x^2 + 12x - 1500 = 0 \quad (3)$$

(c) Hence calculate the total perimeter of the target area, $PQRS$, giving your answer to the nearest metre.

(3)

(Total for Question 9 is 7 marks)

Q32 PAST PAPER Pearson IAL WMA11 · June 2026 · Q7

P1 · TOPIC 10 7 marks

7.

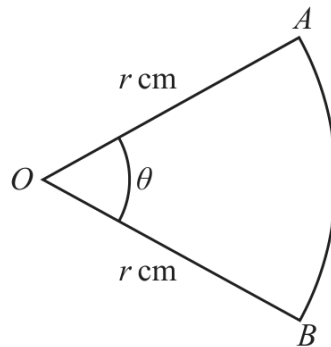
Diagram **NOT**
accurately drawn**Figure 3**

Figure 3 shows a sector AOB of a circle, centre O .

The radius OA of the sector is r cm and angle AOB is θ radians.

Given that

- the **area** of sector AOB is 1 cm^2
- the **perimeter** of sector AOB is 8 cm

(a) write down two different equations involving r and θ .

(2)

Given that $r > 2$

(b) solve the two simultaneous equations in part (a) to find the exact value of r and the exact value of θ .

Write each answer in the form $a + b\sqrt{c}$ where a , b and c are integers.

(Solutions relying on calculator technology are not acceptable.)

(5)

Question 7 continued**(Total for Question 7 is 7 marks)**

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01 Arc & sector** $s = r\theta$, $A = \frac{1}{2}r^2\theta$. Major sector: $2\pi - \theta$. RAD mode.
- 02 Triangles** $\frac{1}{2}ab \sin C$, sine rule, cosine rule. Segment = $\frac{1}{2}r^2(\theta - \sin \theta)$.
- 03 Composite** Name the pieces → missing lengths → add areas, walk the outside.
- 04 Equations** Two facts → two equations in r, θ → substitute.

BOTTOM LINE Radians only → break the shape into sectors and triangles → full accuracy to the last line.

Answers

Radian Measure • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • RADIANS, ARCS AND SECTORS

- Q1** Dr Eslam • Arc and sector arc 4π ; area 12π ; perimeter $12 + 4\pi$
- Q2** Dr Eslam • Find θ $\theta = 1.6$
- Q3** Practice • Easy (i) $\frac{\pi}{3}$ (ii) $\frac{\sqrt{2}}{2}$ (iii) 0 (iv) 150° (v) 1
- Q4** Practice • Easy (i) π cm (ii) 2π cm²
- Q5** Practice • Medium area $\frac{81\pi}{10} \approx 25.4$ cm²; perimeter $18 + \frac{9\pi}{5} \approx 23.7$ cm
- Q6** Pearson IAL WMA11 • October 2019 • Q1 (a) $r = 2\sqrt{6}$ (b) $\frac{13}{2}\sqrt{6}$ cm
- Q7** Pearson IAL WMA11 • June 2022 • Q8 (b) 603 cm (c) 4730 cm² (0.473 m²)
- Q8** Pearson IAL WMA11 • October 2024 • Q5 (a) 16 km (c) $AP = 4.7$ km

IDEA 02 • TRIANGLES AND SEGMENTS

- Q9** Dr Eslam • Segment $\frac{1}{2}(16)(1.2 - \sin 1.2) = 2.14 \text{ cm}^2$
- Q10** Practice • Easy (a) 4.66 cm^2 (b) $1.5\pi = 4.71 \text{ cm}^2$ (c) 0.0536 cm^2
- Q11** Practice • Medium (a)(i) 17.5 (ii) 13.6 (iii) 3.91 cm^2 (b)(i) 6.48 cm (ii) 17.3 cm
- Q12** Pearson IAL WMA11 • January 2021 • Q5 (a) 1.805 (b) 26.9 m (c) 113.7 m^2
- Q13** Pearson IAL WMA11 • October 2021 • Q7 (b) 5.6 m (c) 1.6 m^2
- Q14** Pearson IAL WMA11 • January 2022 • Q5 (b) 3.4 m^2 (c) 9.8 m
- Q15** Pearson IAL WMA11 • October 2022 • Q8 (a) 6π , $6 + 4\pi$ (b) $\frac{9\sqrt{3}}{4}$ (d) 43 m^2
- Q16** Pearson IAL WMA11 • January 2023 • Q6 (a) 5.77 (b) 0.37 (c) 53.1 m^2 (d) 30.9 m
- Q17** Pearson IAL WMA11 • June 2024 • Q5 (a) 23.4 m^2 (b) 1.54 (c) 60.9 m^2
- Q18** Pearson IAL WMA11 • January 2025 • Q8 (a) 2.034 (b) 1.78 cm^2 (d) 9 cm
- Q19** Pearson IAL WMA11 • June 2025 • Q6 (b) 622 m^2 (c) 94.1 m
- Q20** Pearson IAL WMA11 • January 2026 • Q5 (b) 21.9 cm^2 (c) 21.7 cm

IDEA 03 • COMPOSITE SHAPES

- Q21** Dr Eslam • Walk the outside $P = 6 + 8 + 3\pi \approx 23.4$ cm; $A = 24 + 4.5\pi \approx 38.1$ cm²
- Q22** Pearson IAL WMA11 • January 2020 • Q4 (a) $\cos^{-1} \frac{4}{12} = 1.231$ (b) 292.2 m² (c) 79.8 m
- Q23** Pearson IAL WMA11 • June 2021 • Q7 (a) 12 (c) 30 (d) $\frac{950}{3}$
- Q24** Pearson IAL WMA11 • June 2023 • Q5 (b) 18.5 m (c) 15.7 m²
- Q25** Pearson IAL WMA11 • January 2024 • Q8 (b) 56.0 m (c) 186 m²
- Q26** Pearson IAL WMA11 • October 2025 • Q7 (b) 211 m² (c) 57 m

IDEA 04 • EQUATIONS IN r AND θ

- Q27** Dr Eslam • Two facts, two equations $r\theta = 20 - 2r, \frac{1}{2}r(20 - 2r) = 16 \Rightarrow r^2 - 10r + 16 = 0:$
 $r = 2$ or 8
- Q28** Practice • Easy $r = 10, \theta = 0.5$
- Q29** Practice • Easy $r = 4$ cm, $\theta = \pi$
- Q30** Practice • Hard $a = 6$
- Q31** Pearson IAL WMA11 • October 2023 • Q9 (c) $x = 16.2$; perimeter 43 m
- Q32** Pearson IAL WMA11 • June 2026 • Q7 (a) $\frac{1}{2}r^2\theta = 1, 2r + r\theta = 8$ (b) $r = 2 + \sqrt{3}, \theta = 14 - 8\sqrt{3}$